5.1 Condition: change denominator

Definition and results

For $P(A) > 0$, the probability of B knowing A is $P_A(B) = \dfrac{P(A \cap B)}{P(A)}$. In a table It is the effective quotient of $A \cap B$ A. In general $P_A(B) \neq P_B(A)$.

Status and justification. Definition. $P(A)$, we get the rule $P(A \cap B) = P(A)P_A(B)$ The condition $P(A) > 0$ is indispensable.

Example solved. Out of 100 people, 40 belong to A, 30 to B and 12 to both. $P_A(B) = \dfrac{12}{40} =$ 0,3, while $P_B(A) = \dfrac{12}{30} = 0{,}4$. $P(A \cap B) = 0{,}12$.

Point of vigilance

"Among the A" sets the denominator. A conditional probability is not an intersection.

Immediate application. Out of 60 objects, 20 are red; among the red, 5 are damaged. $P(R \cap D)$ and $P_R(D)$.

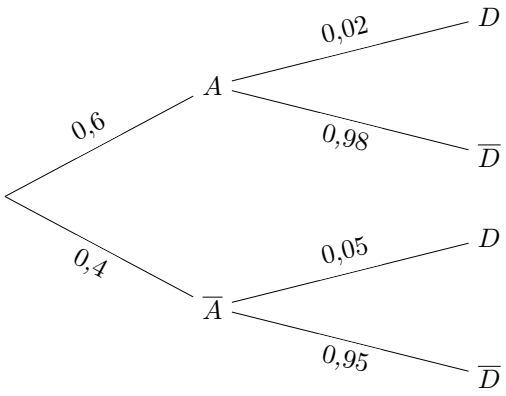
5.2 Trees, partitions and total probabilities

Definition and results

A partition $A_1, \ldots, A_k$ is a family of disjointed events whose meeting is the universe. If their probabilities are positive, $P(B) = \sum_i P(A_i)P_{A_i}(B)$ On a tree, probabilities from the same node have a sum of 1; the probabilities of a path multiply; the Disjointed paths meet by adding together.

Status and justification. B is the separate meeting of the $B \cap A_i$. By additivity and by definition of conditional probabilities, $P(B) = \sum_i P(B \cap A_i) = \sum_i P(A_i)P_{A_i}(B)$.

Example solved. Two machines provide 60% and 40% of production; their default rates 2 % and 5 %. $P(D) = 0{,}6 \times 0{,}02 + 0{,}4 \times 0{,}05 = 0{,}032$. Among the defects, the first machine Representative $\dfrac{0{,}012}{0{,}032} = \dfrac{3}{8}$.



A is the machine that provides 60 percent of the production and D a defect. For example, $0{,}02 = P_A(D)$: the probabilities of the second level are conditional.

Point of vigilance

The simple average of the rates is correct only if the weight of the branches is equal.

Immediate application. $P(A) = \frac{1}{3}$, $P_A(B) = \frac{3}{4}$, $P_{\overline{A}}(B) = \frac{1}{4}$. Calculate $P(B)$.

5.3 Independence and Incompatibility

Definition and results

A and B are independent if $P(A \cap B) = P(A)P(B)$ If $P(A) > 0$, this is equivalent to $P_A(B) = P(B)$ They are incompatible if $A \cap B = \varnothing$. Two events of probability strictly positive and incompatible are not independent.

Status and justification. The conditional criterion results from a division by $P(A) > 0$. For events incompatible with positive probabilities, the left member is 0 and the product of right is strictly positive.

Example solved. On a balanced dice, A: even result, B: result at least 5. $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$, $P(A \cap B) = \frac{1}{6}$: A and "unfair result" are incompatible but not independent.

Point of vigilance

An independence in a model must be explicitly justified or laid down; "two events different" is not enough.

Immediate application. We give $P(A) = 0{,}4$, $P(B) = 0{,}5$, $P(A \cap B) = 0{,}18$. Are they independent?

5.4 Successive Tests and Bernoulli

Definition and results

A Bernoulli test has two results: success, probability p, and failure, probability 1 $-p$. In a repetition of independent and identical tests, with $n \leq 4$ here, probabilities branches remain p and 1 $-\,p$. A path of success is likely $p^k(1-p)^{n-k}$; on counts the paths that realize the event, without general theory of binomial distribution.

Status and justification. The probability of a path is a product, by independence. corresponding to an event are disjointed: their probabilities add up.

Example solved. For three tests with $p = \frac{1}{2}$, exactly two successes correspond to SSE, SES, ESS. Each path has a probability $\frac{1}{8}$, therefore total $\frac{3}{8}$. At least one success is likely $1 - (\frac{1}{2})^3 = \frac{7}{8}$.

Point of vigilance

Without a return to a ballot box, the probability generally changes after the first draw: Independent model can be false.

Immediate application. Four independent probability tests $p = \frac{1}{3}$. Calculate probability four failures and then at least one success.

5.5 Simulate and discuss the model

Definition and results

A pseudo-random simulation produces frequencies that fluctuate around a probability for a uniform point in the square [0; 1] ^{2}, the proportion checking $y \leq x^2$ estimates under the parabola; the proportion checking $x^2 + y^2 \leq 1$, multiplied by 4, estimates π. The repeats must use new prints.

Status and justification. Accepted and experienced simulation principle. is an interpretation of the uniform drawing; integration is not a prerequisite. Doesn't show an exact value.

Example solved. Among 10,000 points, if 7840 are in the quarter disk, the estimate of π value $\frac{4 \times 7840}{10000} = 3{,}136$. Two tests of the same size usually give different values.

Point of vigilance

Fixing a seed makes it possible to reproduce a test, not to remove the alea from the model or from the ensure accuracy.

Immediate application. A simulation of 2000 points gives 675 points under $y = x^2$. What Does she think so?